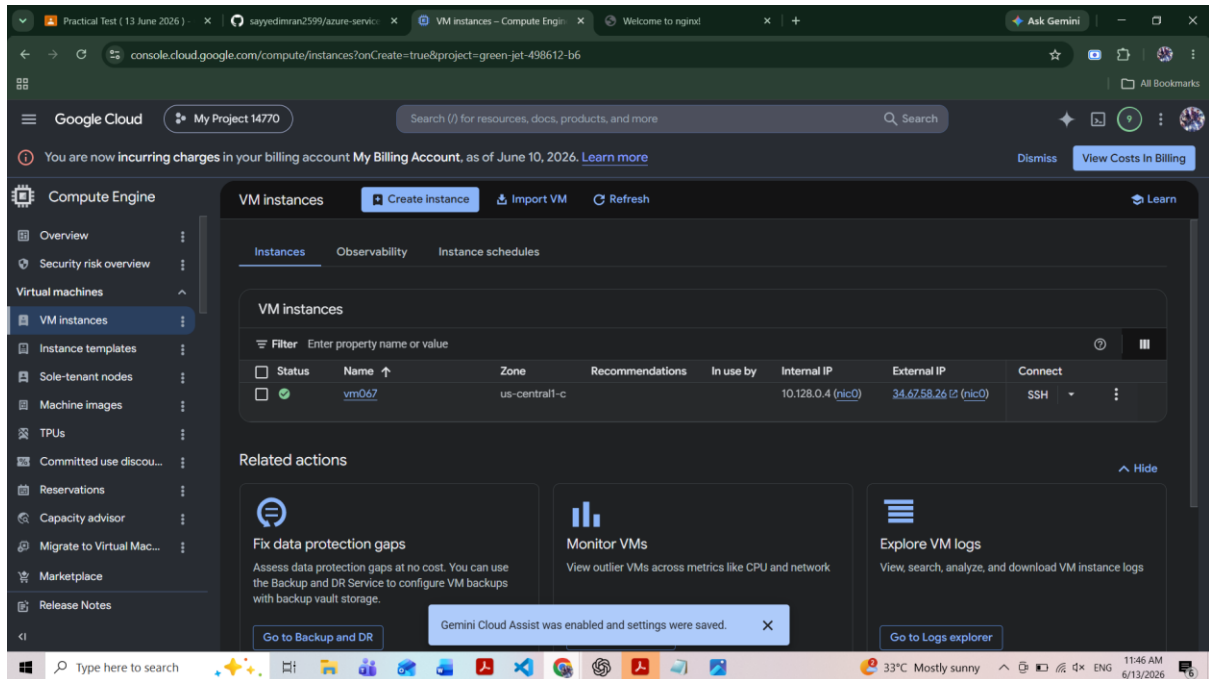
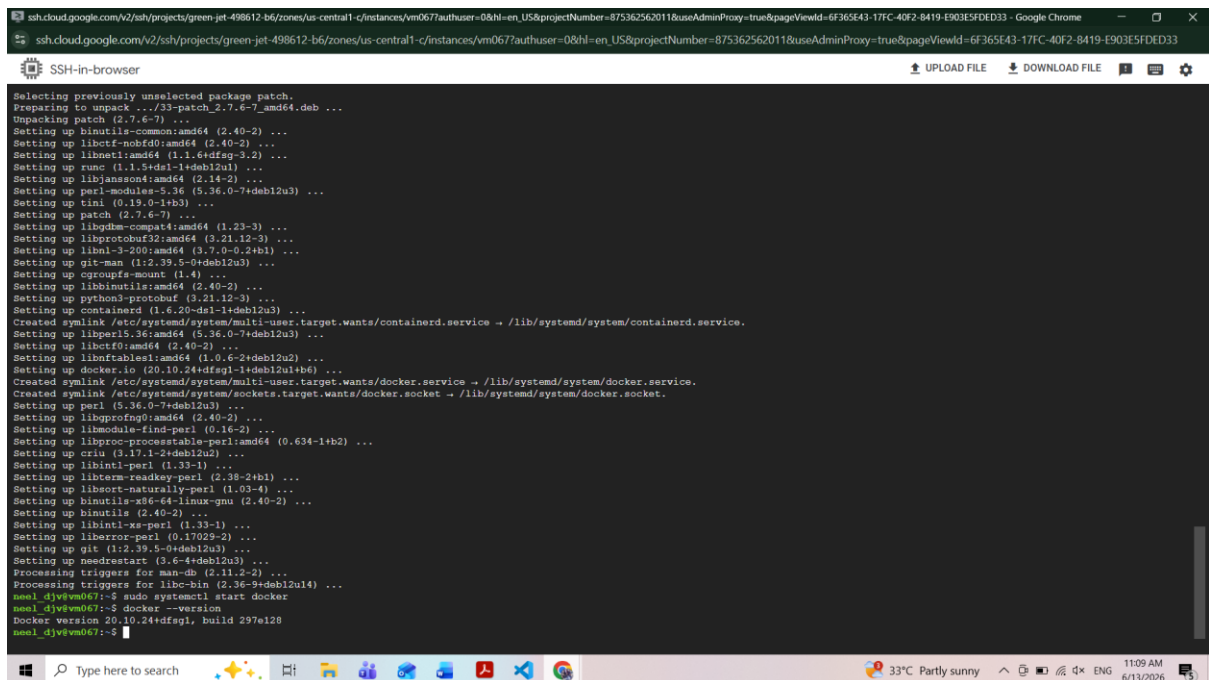


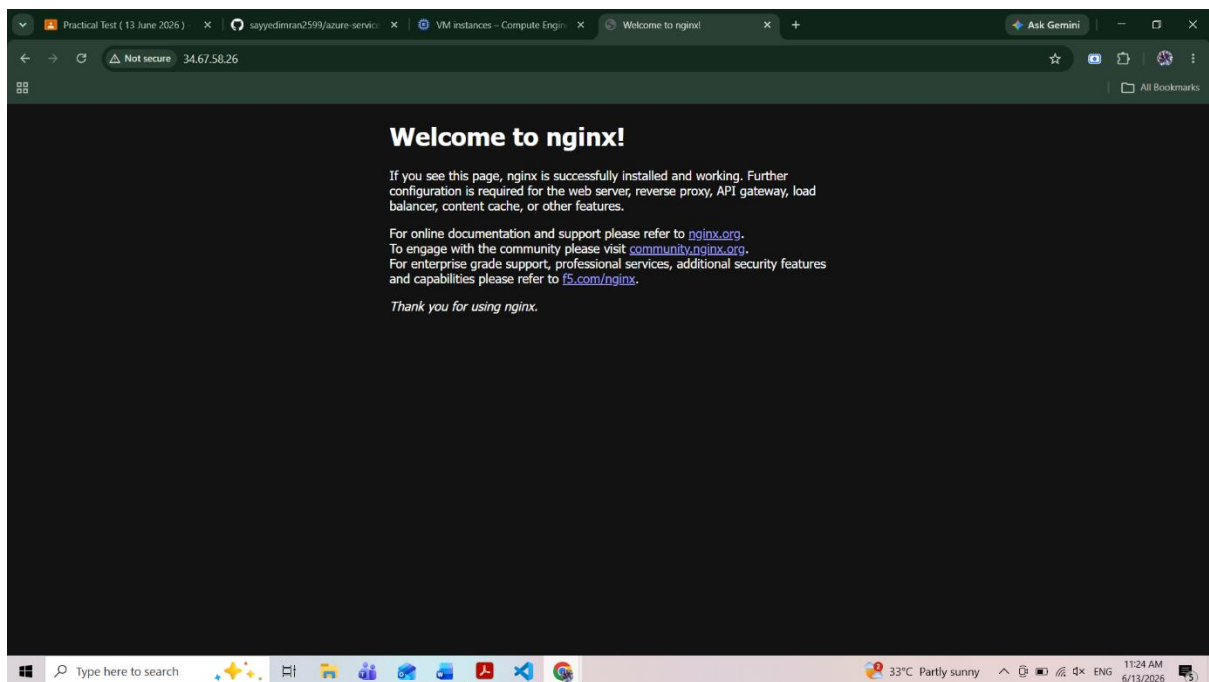
BATCH 1 CMPN



Q.1)



```
ssh.cloud.google.com/v2/ssh/projects/green-jet-498612-b6/zones/us-central1-c/instances/vm067?authuser=0&hl=en_US&projectNumber=875362562011&useAdminProxy=true&pageViewId=6f365E43-17FC-40F2-8419-E903E5FDED33 - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/green-jet-498612-b6/zones/us-central1-c/instances/vm067?authuser=0&hl=en_US&projectNumber=875362562011&useAdminProxy=true&pageViewId=6f365E43-17FC-40F2-8419-E903E5FDED33
SSH-in-browser
UPLOAD FILE
DOWNLOAD FILE
Linux vm067 6.1.0-49-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.174-1 (2026-05-26) x86_64
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jun 13 05:44:08 2026 from 35.235.249.250
meel_dj@vm067:~$ # Pull the Nginx image
sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:608a100c71451bf5b773e99083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
meel_dj@vm067:~$
```



Q.2)

```
ssh.cloud.google.com/v2/ssh/projects/green-jet-498612-b6/zones/us-central1-c/instances/vm067/authuser=0&hl=en_US&projectNumber=875362562011&useAdminProxy=true&pageViewId=6f365e43-17fc-40f2-8419-E903E5FDED33 - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/green-jet-498612-b6/zones/us-central1-c/instances/vm067/authuser=0&hl=en_US&projectNumber=875362562011&useAdminProxy=true&pageViewId=6f365e43-17fc-40f2-8419-E903E5FDED33

SSH-in-browser

Linux vm067 6.1.0-49-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.174-1 (2026-05-26) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jun 13 05:44:08 2026 from 35.235.249.250
neel_dj@vm067:~$ # Pull the Nginx image
sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:608a100c71651bf5b773c89083b4alad7ef4b2bd05d7a7e552271e03123692ad
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
neel_dj@vm067:~$ sudo docker run --name my-nginx -p 80:80 -d nginx
docker: Error response from daemon: Conflict. The container name "/my-nginx" is already in use by container "c9d5f62c72e7eb1925bc9a0528dc497d12a1b717794f585f9cb156e130c5941". You have to remov
e (or rename) that container to be able to reuse that name.
See 'docker run --help'.
neel_dj@vm067:~$ docker volume create studentdata
studentdata
neel_dj@vm067:~$ sudo docker run -it --name ubuntu_student -v studentdata:/data ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
6f5c5aa4e145: Pull complete
1c24335ddd46: Pull complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@5b942b5b8cd8:/#
```

```
ssh.cloud.google.com/v2/ssh/projects/green-jet-498612-b6/zones/us-central1-c/instances/vm067/authuser=0&hl=en_US&projectNumber=875362562011&useAdminProxy=true&pageViewId=6f365e43-17fc-40f2-8419-E903E5FDED33 - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/green-jet-498612-b6/zones/us-central1-c/instances/vm067/authuser=0&hl=en_US&projectNumber=875362562011&useAdminProxy=true&pageViewId=6f365e43-17fc-40f2-8419-E903E5FDED33

SSH-in-browser

Linux vm067 6.1.0-49-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.174-1 (2026-05-26) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jun 13 05:44:08 2026 from 35.235.249.250
neel_dj@vm067:~$ # Pull the Nginx image
sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:608a100c71651bf5b773c89083b4alad7ef4b2bd05d7a7e552271e03123692ad
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
neel_dj@vm067:~$ sudo docker run --name my-nginx -p 80:80 -d nginx
docker: Error response from daemon: Conflict. The container name "/my-nginx" is already in use by container "c9d5f62c72e7eb1925bc9a0528dc497d12a1b717794f585f9cb156e130c5941". You have to remov
e (or rename) that container to be able to reuse that name.
See 'docker run --help'.
neel_dj@vm067:~$ docker volume create studentdata
studentdata
neel_dj@vm067:~$ sudo docker run -it --name ubuntu_student -v studentdata:/data ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
6f5c5aa4e145: Pull complete
1c24335ddd46: Pull complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@5b942b5b8cd8:/# touch /data/info.txt
root@5b942b5b8cd8:/# echo "Name: Neel Vaidya" > /data/info.txt
root@5b942b5b8cd8:/# echo "Roll No: <Your_Roll_Number>" >> /data/info.txt
root@5b942b5b8cd8:/# echo "Name: Neel Vaidya" > /data/info.txt
root@5b942b5b8cd8:/# echo "Roll No: CS22045" >> /data/info.txt
root@5b942b5b8cd8:/# cat /data/info.txt
Name: Neel Vaidya
Roll No: CS22045
root@5b942b5b8cd8:/# docker run -it --rm -v studentdata:/data ubuntu cat /data/info.txt
bash: docker: command not found
root@5b942b5b8cd8:/# sudo docker run -it --rm -v studentdata:/data ubuntu cat /data/info.txt
bash: sudo: command not found
root@5b942b5b8cd8:/#
```

Q.3)

```
ssh.cloud.google.com/v2/ssh/projects/green-jet-498612-b6/zones/us-central1-c/instances/vm067/authuser-08hl-en_US&projectNumber=875362562011&useAdminProxy=true&pageViewId=6f365e43-17fc-40f2-8419-E903E5FDED33 - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/green-jet-498612-b6/zones/us-central1-c/instances/vm067/authuser-08hl-en_US&projectNumber=875362562011&useAdminProxy=true&pageViewId=6f365e43-17fc-40f2-8419-E903E5FDED33

SSH-in-browser

the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
last login: Sat Jun 13 05:44:09 2026 from 35.235.249.250
neel@djv6vm67:~$ # Pull the Nginx image
sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:608a100c71451bf5b773c89083b4alad7ef4b2bd05d7a7e552271e03123692ad
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
neel@djv6vm67:~$ sudo docker run --name my-nginx -p 80:80 -d nginx
docker: Error response from daemon: Conflict. The container name "/my-nginx" is already in use by container "c9d5f62c72e7eb1925bc9a0528dcd497d12alb717794f985f9cb156e130c5941". You have to remov
e (or rename) that container to be able to reuse that name.
See 'docker run --help'.
neel@djv6vm67:~$ docker volume create studentdata
studentdata
neel@djv6vm67:~$ docker run -it --name ubuntu_student -v studentdata:/data ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
6f5c5aa4e145: Pull complete
1c24335ddd46: Pull complete
Digest: sha256:f3d28607dd78734bb7f71f117f3c6706c666b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@85b942b5b8cd:/# touch /data/info.txt
root@85b942b5b8cd:/# echo "Name: Neel Vaidya" > /data/info.txt
root@85b942b5b8cd:/# echo "Roll No: <Your_Roll_Number>" >> /data/info.txt
root@85b942b5b8cd:/# echo "Name: Neel Vaidya" > /data/info.txt
root@85b942b5b8cd:/# echo "Roll No: CS22045" >> /data/info.txt
root@85b942b5b8cd:/# cat /data/info.txt
Name: Neel Vaidya
Roll No: CS22045
root@85b942b5b8cd:/# docker run -it --rm -v studentdata:/data ubuntu cat /data/info.txt
bash: docker: command not found
root@85b942b5b8cd:/# sudo docker run -it --rm -v studentdata:/data ubuntu cat /data/info.txt
bash: sudo: command not found
root@85b942b5b8cd:/# FROM ubuntu:latest

CMD ["echo", "Cloud Computing Practical Exam Completed Successfully"]
bash: FROM: command not found
bash: CMD: command not found
root@85b942b5b8cd:/# nano Dockerfile
bash: nano: command not found
root@85b942b5b8cd:/#
```

```
ssh.cloud.google.com/v2/ssh/projects/green-jet-498612-b6/zones/us-central1-c/instances/vm067/authuser-08hl-en_US&projectNumber=875362562011&useAdminProxy=true&pageViewId=6f365e43-17fc-40f2-8419-E903E5FDED33 - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/green-jet-498612-b6/zones/us-central1-c/instances/vm067/authuser-08hl-en_US&projectNumber=875362562011&useAdminProxy=true&pageViewId=6f365e43-17fc-40f2-8419-E903E5FDED33

SSH-in-browser

root@85b942b5b8cd:/# echo "Name: Neel Vaidya" > /data/info.txt
root@85b942b5b8cd:/# echo "Roll No: <Your_Roll_Number>" >> /data/info.txt
root@85b942b5b8cd:/# echo "Name: Neel Vaidya" > /data/info.txt
root@85b942b5b8cd:/# echo "Roll No: CS22045" >> /data/info.txt
root@85b942b5b8cd:/# cat /data/info.txt
Name: Neel Vaidya
Roll No: CS22045
root@85b942b5b8cd:/# docker run -it --rm -v studentdata:/data ubuntu cat /data/info.txt
bash: docker: command not found
root@85b942b5b8cd:/# sudo docker run -it --rm -v studentdata:/data ubuntu cat /data/info.txt
bash: sudo: command not found
root@85b942b5b8cd:/# FROM ubuntu:latest

CMD ["echo", "Cloud Computing Practical Exam Completed Successfully"]
bash: FROM: command not found
bash: CMD: command not found
root@85b942b5b8cd:/# nano Dockerfile
bash: nano: command not found
root@85b942b5b8cd:/# docker run ...
bash: docker: command not found
root@85b942b5b8cd:/# sudo docker run ...
bash: sudo: command not found
root@85b942b5b8cd:/# exit
exit
neel@djv6vm67:~$ nano Dockerfile
neel@djv6vm67:~$ docker build -t practical-exam-image .
Sending build context to Docker daemon 12.29kB
Step 1/2 : FROM ubuntu:latest
--> 30ba4450a4d
Step 2/2 : CMD ["echo", "Cloud Computing Practical Exam Completed Successfully"]
--> Running in a3e9c0e0aacc
Removing intermediate container a3e9c0e0aacc
--> 421315a38207
Successfully built 421315a38207
Successfully tagged practical-exam-image:latest
neel@djv6vm67:~$ docker run practical-exam-image
Cloud Computing Practical Exam Completed Successfully
neel@djv6vm67:~$ docker images
REPOSITORY          TAG         IMAGE ID      CREATED       SIZE
practical-exam-image latest      421315a38207  25 seconds ago 100MB
nginx                latest     936fe4290c8f  2 days ago   161MB
ubuntu              latest     30ba4450a6d   7 weeks ago  100MB
hello-world         latest     e2ac70e7319a  2 months ago  10.1kB
neel@djv6vm67:~$ docker run --rm practical-exam-image
Cloud Computing Practical Exam Completed Successfully
neel@djv6vm67:~$
```

## BONUS QUESTION

### 1. Difference between a Docker Image and a Docker Container

- A Docker Image is a read-only template that contains the application code, libraries, dependencies, and configuration required to run an application.
- A Docker Container is a running instance of a Docker image. Containers are created from images and can be started, stopped, modified, or deleted.

## 2. Why are Docker Volumes used?

- Docker Volumes are used to store and manage persistent data outside the container's filesystem.
- They ensure that data is not lost when a container is stopped, removed, or recreated, and allow multiple containers to share the same data.